

CNW 2140 Computer Networks

Course syllabus

School of Computing, Tarnhollow University

Autumn term 2025. Version 2.0, issued Monday 1 September 2025.

1. Teaching team

Lecturer: Dr. Oleska Brantigern, room HV 3.07. Office hours are Monday 11:00 to 12:30, no appointment needed.

Teaching assistant: Corwin Ashgrove-Lund, who runs the labs and marks the lab reports. Lab drop-in is Thursday 16:00 to 17:00 in room HV 1.22.

Use the course forum for anything other students might also want to know. Email is for personal matters only, and is read within two working days.

2. Meetings

Lectures are on Monday and Wednesday, 14:00 to 15:30, in the Hale-Voss Building, room HV 1.05. The lab session is Friday 10:00 to 12:00 in HV 1.22. There are 22 lectures. Teaching starts on Monday 8 September 2025 and the last lecture is on Wednesday 26 November 2025.

3. What the course covers

The course follows a packet from the wire to the application and back. In order, the first half of the term covers: the layered model; links, frames and error detection; switching and Ethernet; sockets and the transport layer; DNS; HTTP; TCP congestion control; TCP reliability and retransmission; IP addressing and subnets; routing inside and between networks; and TLS. The second half covers wireless networks, network security and data-centre networking. A dated lecture plan is posted on the course site in the first week.

4. Assessment

- Labs: 15% of the course mark. Five labs, 3% each.
- Midterm exam: 25% of the course mark. Wednesday 29 October 2025, in the lecture slot, 80 minutes. Covers lectures 1 to 8.
- Group project: 10% of the course mark. Groups of three. Report due Friday 28 November 2025.
- Final exam: 50% of the course mark. Thursday 11 December 2025, 14:00 to 16:00, Examination Hall 1. Covers the whole course.

To pass the course you need an overall mark of at least 50.

Materials: both exams are closed book. Calculators are not needed and not allowed.

5. Labs and deadlines

Lab reports are submitted through the course site by 17:00 on the due date.

- Lab 1, packet capture warm-up: due Friday 19 September 2025

- Lab 2, a socket echo server: due Friday 3 October 2025
- Lab 3, TCP trace analysis: due Friday 17 October 2025
- Lab 4, build a DNS resolver: due Friday 7 November 2025
- Lab 5, an HTTP cache: due Friday 21 November 2025

Late policy: a late lab report loses 5 marks out of 100 for each day or part of a day after the deadline, for up to 5 days. After 5 days it scores zero. Extensions are granted only through the School office, not by the lecturer or the teaching assistant.

Regrade requests must reach the teaching assistant within 7 days of the mark being released.

6. Reading

The course text is Halloway and Minch, *Networks from the Wire Up*, 2nd edition. The library holds eight copies on short loan.

7. Working together and working honestly

Lab reports are individual work. You may discuss approaches with other students, but the captures, the code and the write-up must be your own. The group project is marked as a group. Suspected plagiarism is referred to the School's integrity officer, and the lecturer has no discretion in that.

If you have a learning support plan, send it to the lecturer in the first two weeks of term so that exam arrangements can be made in time.